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Digitally controlled delay line gain calibration using error injection

Abstract

The techniques described herein relate to digitally controlled delay line gain calibration using error injection. An example apparatus includes a digitally controlled delay line (DCDL) with a DCDL output and configured to: receive a clock signal to be output from a voltage-controlled oscillator, and delay the clock signal to generate a first delayed clock signal. The apparatus further includes an error injection circuit with a first error injection input and a second error injection input, the first error injection input coupled to the DCDL output. The apparatus additionally includes a phase controller with a phase controller output coupled to the second error injection input, the phase controller configured to, in response to a pseudorandom binary sequence signal, instruct the error injection circuit to generate a second delayed clock signal based on a delay of the first delayed clock signal.

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Background/Summary

RELATED APPLICATION (1) This patent claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/383,923, titled “DIGITALLY CONTROLLED DELAY LINE GAIN CALIBRATION USING ERROR INJECTION,” filed on Nov. 16, 2022, which is hereby incorporated by reference herein in its entirety.

FIELD

(1) The techniques described herein relate generally to frequency synthesizers and, more particularly, to digitally controlled delay line gain calibration using error injection.

BACKGROUND

(2) Receivers, such as wireline or wireless signal receivers, are devices that may receive

electromagnetic signals. The electromagnetic signals may include high-frequency and low-frequency signal components. Some wireline signal receivers may use frequency synthesizers to generate a waveform at a frequency determined by analog or digital circuits. For instance, a frequency synthesizer may be an electronic device that uses an oscillator to generate a signal with a specific frequency or within a pre-set frequency range. Operation of some such frequency synthesizers may be adversely affected by component and/or system noise.

SUMMARY OF THE DISCLOSURE

(3) Some aspects relate to an apparatus for calibrating a phase-locked loop, comprising a digitally controlled delay line (DCDL) with a DCDL output and configured to: receive a clock signal to be output from a voltage-controlled oscillator; and delay the clock signal to generate a first delayed clock signal; an error injection circuit with a first error injection input and a second error injection input, the first error injection input coupled to the DCDL output; and a phase controller with a phase controller output coupled to the second error injection input, the phase controller configured to, in response to a pseudorandom binary sequence signal, instruct the error injection circuit to generate a second delayed clock signal based on a delay of the first delayed clock signal.

(4) Some aspects relate to another apparatus for calibrating a phase-locked loop, comprising a phase detector with a first phase detector input and a second phase detector input and configured to determine a phase difference between a first clock signal associated with an input path and a second clock signal associated with a feedback path; a first error injection circuit with a first error input and a first error output, the first error input configured to receive an input clock signal, the first error output coupled to the first phase detector input, and the first error injection circuit is configured to delay the input clock signal by a first delay to generate the first clock signal; and a second error injection circuit with a second error input and a second error output, the second error input configured to receive a feedback clock signal, the second error output coupled to the second phase detector input, and the second error injection circuit is configured to delay the feedback clock signal by a second delay to generate the second clock signal.

(5) Some aspects relate to yet another apparatus for calibrating a phase-locked loop, comprising a digitally controlled delay line (DCDL) with a DCDL output and configured to: receive a clock signal, the clock signal to be output from a voltage-controlled oscillator; and delay the clock signal to generate a first delayed clock signal; an error injection circuit with a first error injection input and a second error injection input, the first error injection input coupled to the DCDL output; and a phase controller with a phase controller output coupled to the second error injection input, the phase controller configured to, in response to a pseudorandom binary sequence signal, instruct the error injection circuit to generate a second delayed signal based on a delay of the first delayed clock signal.

(6) Some aspects relate to a method comprising receiving an input clock signal associated with a phase-locked loop; calibrating a first error signal based on half a period of a voltage-controlled oscillator; identifying an error margin associated with the first error signal to generate a first error value and a second error value; injecting a second error signal corresponding to the first error value in the phase-locked loop; injecting a third error signal corresponding to the second error value in the phase-locked loop; and calibrating a delay of the input clock signal based on at least one of the second error signal or the third error signal.

(7) The foregoing summary is not intended to be limiting. Moreover, various aspects of the present disclosure may be implemented alone or in combination with other aspects.

Description

BRIEF DESCRIPTION OF FIGURES

(1) In the drawings, each identical or nearly identical component that is illustrated in various

figures is represented by a like reference character. For purposes of clarity, not every component may be labeled in every drawing. The drawings are not necessarily drawn to scale, with emphasis instead being placed on illustrating various aspects of the techniques and devices described herein.

(2) FIG. 1 depicts a schematic illustration of an example phase-locked loop including error injection circuits to calibrate a digitally controlled delay line of the phase-locked loop.

(3) FIG. 2 is a plot depicting example operation of the error injection circuits of FIG. 1.

(4) FIG. 3 depicts an example implementation of a first error injection circuit and a second error injection circuit.

(5) FIG. 4 depicts another example implementation of a first error injection circuit and a second error injection circuit.

(6) FIG. 5 is a plot of an example calibration operation of the digitally controlled delay line of FIG. 1.

(7) FIG. 6 depicts an example dead-zone analysis for determination of error values.

(8) FIG. 7 depicts another example dead-zone analysis for determination of error values.

(9) FIG. 8 depicts a combination of the example dead-zone analyses of FIGS. 6 and/or 7 for determination of error values.

(10) FIG. 9 is a flowchart representative of an example process that may be performed and/or implemented using hardware logic or machine-readable instructions that may be executed by processor circuitry to implement the phase-locked loop of FIG. 1.

DETAILED DESCRIPTION

(11) Phase-locked loop (PLL) circuits are used in a wide variety of high frequency applications. Non-limiting examples of high frequency applications include clock clean-up circuits, local oscillators (LOs) for high performance communication links, and ultrafast switching frequency synthesizers. Non-limiting examples of high-performance communication links include wireline communication links, such as Ethernet links, and wireless communication links, such as radiofrequency (RF), radar, and satellite communication links. Some PLL circuits include an oscillator (e.g., a digitally controlled oscillator (DCO), a voltage-controlled oscillator (VCO), a voltage-driven oscillator (VDO)) that adjusts (e.g., constantly adjusts) to match the frequency of an input signal. For example, some such PLL circuits may be used to generate, stabilize, modulate, demodulate, filter, or recover a signal from a communications channel from which the reception of data may be affected by noise associated with the communications channel.

(12) Some PLL circuits may output signals, such as clock signals, which are affected by phase drift over time. Some such PLL circuits use a phase rotator (e.g., phase rotator circuit) to receive a clock phase from the PLL and shift (e.g., modulo shift) the clock phase into a desired phase angle to overcome the phase drift. However, phase rotators can consume a substantial amount of power and area. Additionally, phase rotators can have high non-linearity and limited resolution. In some instances, it can be challenging to calibrate the frequency of the phase rotators.

(13) Some PLL circuits utilize a digitally controlled delay line (DCDL) to overcome the shortcomings of phase rotators. For example, a DCDL (e.g., a DCDL circuit) can consume substantially less power and area than a phase rotator. In some instances, a DCDL has improved linearity and resolution when compared to a phase rotator.

(14) One challenge of using a DCDL to change a phase of a signal output from a PLL is that the DCDL can be a source of phase noise and spurs (e.g., fractional spurs). Such phase noise and spurs can degrade signal quality and overall operation of the PLL, such as by introducing jitter (e.g., deterministic jitter) into the PLL. Another challenge of using a DCDL is that, although more linear than a phase rotator, the DCDL may have non-linear characteristics. Yet another challenge of using a DCDL is gain error. For example, the gain error of a DCDL can be representative of how well and/or closely the slope of an actual transfer function of the DCDL matches and/or overlaps the slope of the ideal transfer function for the DCDL.

(15) There are challenges with calibrating DCDL gain error. One such challenge is that when a PLL

circuit generates a pulse position modulation (PPM) signal, the PPM signal can be substantially attenuated and thereby result in an inaccurate DCDL gain. Another such challenge is when a PLL includes a fractional-N (Frac-N) circuit, the Frac-N circuit may use the same error template as a dual clock compare (DCC) circuit of the PLL and thusly the DCC and DCDL rang errors may be indistinguishable. Yet another such challenge is that coupling of the supply and substrate may cause the error from the DCDL range and the coupling may be indistinguishable.

(16) The inventors have recognized that the aforementioned challenges of using phase rotators in PLL circuits have not been overcome by using DCDLs. To overcome the deficiencies of phase rotators and/or DCDLs in PLL circuits, the inventors have developed technology for digitally controlled delay line calibration using error injection.

(17) Exemplary PLL circuits disclosed herein overcome the aforementioned challenges by including error injection circuits that can intentionally introduce delay signals to shift a signal, such as a clock signal, into a different error detection region. In some disclosed embodiments, a PLL circuit can include a DCDL and at least a first error injection circuit. The DCDL can be configured to delay an output signal, such as an output clock signal from an oscillator (e.g., a voltage-controlled oscillator), by a first delay (e.g., a first delay time, a first delay period, a first delay time period) to generate a delayed output signal. The first error injection circuit can be configured to delay the delayed output signal by a second delay (e.g., a second delay time, a second delay period, a second delay time period) to generate a delayed feedback signal. The example PLL circuit can include a phase detector configured to compare a phase difference between an input signal, such as an input clock signal, and the delayed feedback signal. Beneficially, the second delayed signal can be generated such that the phase difference is reduced.

(18) In some embodiments, the PLL circuit can include a second error injection circuit that can be configured to delay the input signal by a third delay (e.g., a third delay time, a third delay period, a third delay time period) to generate a delayed input signal. The phase detector can be configured to compare a phase difference between the delayed input signal and the delayed feedback signal. Beneficially, the input delayed signal can be generated such that the phase difference is reduced.

(19) In some embodiments, the PLL circuit can include digital logic, which can implement a phase controller, to control at least one of the first error injection circuit or the second error injection circuit. For example, the digital logic can instruct the first error injection circuit to delay the output signal in response to a first pseudorandom binary sequence signal. In some embodiments, the digital logic can instruct the first error injection circuit to pass the output signal without delay in response to a second pseudorandom binary sequence signal.

(20) In some embodiments, the digital logic can instruct the second error injection circuit to delay the input signal in response to a third pseudorandom binary sequence signal. In some embodiments, the digital logic can instruct the second error injection circuit to pass the input signal without delay in response to a fourth pseudorandom binary sequence signal.

(21) Beneficially, at least one of the first error injection circuit or the second error injection circuit can delay respective signals such that a gain error of the DCDL can be reduced. Beneficially, by reducing the DCDL gain error, the phase difference detected by the phase detector can also be reduced, which can improve overall operation of the PLL, improve stability of the output signals generated by the PLL, and reduce sensitivity to noise sources such as phase noise and spurs.

(22) Turning to the figures, the illustrated example of FIG. 1 depicts an example phase-locked loop (PLL) **100**. The PLL **100** is a circuit (e.g., a PLL circuit) that generates an output clock signal **104** (identified by CLKOUT) whose phase is related to a phase of a reference clock signal **102** (identified by REFCLK). For example, the PLL **100** can synchronize and/or lock a phase of the output clock signal **104** with a phase of the reference clock signal **102**. In some embodiments, the PLL **100**, or portion(s) thereof, can implement a frequency synthesizer that produces a range of frequencies from a single fixed oscillator.

(23) In the illustrated example, the reference clock signal **102** is an input signal (e.g., an input clock

signal) that can be transmitted by a transmitter and/or received by a receiver. For example, the PLL **100** can be configured to receive the reference clock signal **102** from a wireline receiver, such as a data communication wireline receiver. Non-limiting examples of wireline receivers include Ethernet interfaces, Peripheral Component Interconnect (PCI) interfaces, Serial Digital Interfaces (SDI), Universal Serial Bus (USB) interfaces, and High-Definition Multimedia Interfaces (HDMI). Alternatively, the PLL **100** can be configured to receive the reference clock signal **102** from a wireless receiver. Non-limiting examples of wireless receivers include Wireless Fidelity (Wi-Fi) receivers, Bluetooth receivers, near-field communication (NFC) receivers, radio-frequency identification (RFID) receivers, and satellite receivers (e.g., beyond-line-of-site (BLOS) satellite receivers, line-of-site (LOS) satellite receivers, etc.).

(24) In some embodiments, the PLL **100** is included in and/or associated with an electronic device. Non-limiting examples of electronic devices include gateways, routers, switches, laptop computers, tablet computers, cellular phones (e.g., smartphones), televisions (e.g., smart televisions), set-top boxes, streaming devices, and wearable devices (e.g., headphones, headsets, smartwatches, smart glasses, etc.). For example, the output clock signal **104** can be provided to additional circuitry, such as a transmitter, a receiver, and/or a programmable processor. Non-limiting examples of programmable processors include central processing units (CPUs), digital signal processors (DSPs), graphics processing units (GPUs), and field programmable gate arrays (FPGAs).

(25) In some embodiments, the reference clock signal **102** is a clock signal whose frequency has been doubled and/or otherwise increased by a frequency doubler (not shown but may be included in the PLL **100**). For example, the reference clock signal **102** can be a doubled reference clock signal. In some embodiments, the frequency doubler is a frequency doubler circuit that can increase the frequency of the reference clock signal **102** to increase a bandwidth of the PLL **100**. In some embodiments, the frequency doubler can be implemented by an oscillator (e.g., a reference oscillator, an oscillator circuit) to increase the frequency of the reference clock signal **102**. In some embodiments, the frequency doubler can be configured to receive the reference clock signal **102** from a receiver (e.g., a wireline receiver, a wireless receiver). Alternatively, the PLL **100** may utilize a different frequency increaser than the frequency doubler to triple, quadruple, etc., the reference clock signal **102**.

(26) The PLL **100** of the illustrated example includes a phase detector **106** (identified by PD and may also be referred to as a phase comparator or mixer) to compare a first phase of the reference clock signal **102**, or delayed version(s) thereof, and a second phase of a feedback clock signal **108** (identified by FBCLK). The feedback clock signal **108** can be a delayed instance of the output clock signal **104**. In some embodiments, the PD **106** can be a phase detector circuit that can generate and/or output a voltage according to a phase difference of the first and second phases. In some embodiments, the voltage can be an error signal **107** that is representative of an error that is detected between the phases of the reference clock signal **102** and the feedback clock signal **108**. The PD **106** of the illustrated example has a first input (e.g., a first detector input, a first phase detector input) configured to receive the reference clock signal **102**. For example, the first input can be coupled to an output (e.g., a doubler output, a frequency doubler output) of a frequency doubler. For example, the PD **106** and the frequency doubler can be coupled together through one or more electrical connections. Non-limiting examples of electrical connections include opto-isolators, pads, traces, wires, and vias.

(27) The PLL **100** of this example includes a loop filter **110** (identified by LF). In some embodiments, the LF **110** is a loop filter circuit that converts the output of the PD **106** into a control signal (e.g., a control voltage) for a voltage-controlled oscillator **112** (identified by VCO) of the PLL **100**. For example, the PD **106** can be implemented by one or more charge pumps that can output a current representative of the detected error. In some such embodiments, the LF **110** can convert the current from the one or more charge pumps to the control voltage for the VCO **112**. Alternatively, the PD **106** may output a voltage representative of the detected error. For example,

the PD **106** can be implemented by one or more analog circuits configured to output a voltage representative of the detected error. In some embodiments, the LF **110** can filter out and/or attenuate noise coming from the reference clock signal **102** to the control voltage. The LF **110** of the illustrated example has an input (e.g., a filter input, a loop filter input) coupled to an output (e.g., a detector output, a phase detector output) of the PD **106**.

(28) The PLL **100** of this example includes the VCO **112** to generate and/or output the output clock signal **104** according to the control voltage output from the LF **110**. In some embodiments, the VCO **112** is a VCO circuit that generates and/or outputs the output clock signal **104**. For example, the VCO **112** can be implemented by one or more analog circuits configured to generate and/or output the output clock signal **104**. In some embodiments, the output clock signal **104** is a signal (e.g., a sinusoidal signal) whose frequency closely matches (e.g., matches within 1%, 5%, etc.) of the center frequency provided by the LF **110**. The VCO **112** of this example has an input (e.g., an oscillator input, a VCO input) coupled to an output (e.g., a filter output, a loop filter output) of the LF **110**.

(29) In the illustrated example, the PLL **100** includes a multi-modulus divider **114** (identified by MMD) to divide and/or reduce a frequency of the output clock signal **104**. Additionally or alternatively, a pre-division ratio may be included in the PLL **100** prior to the MMD **114**. In some embodiments, the MMD **114** can be an MMD circuit that divides a frequency of the output clock signal **104** by a divisor (e.g., 2, 3, 4, etc.) to generate a divided clock signal. For example, the MMD **114** can be implemented using one or more analog and/or digital circuits configured to divide the frequency of the output clock signal **104**. In some embodiments, the MMD **114** can delay the divided clock signal by a time delay (e.g., a time duration, a time period) in a time delay range to generate a delayed clock signal. For example, time delays in the time delay range can range from zero time delay to a time delay up to at least a period of the VCO **112** (e.g., T.sub.VCO). Any other time delay range may be utilized. The MMD **114** of this example has an input (e.g., a divider input, an MMD input) coupled to an output (e.g., an oscillator output) of the VCO **112**.

(30) The PLL **100** of the illustrated example includes a digitally controlled delay line (DCDL) **116** to delay the output, such as a divided clock signal, from the MMD **114**, by a time delay in a time delay range to generate the feedback clock signal **108**. In some embodiments, the DCDL **116** can cause a reduction of a difference (e.g., a difference in phases) of the reference clock signal **102** and the feedback clock signal **108**. For example, the difference can be representative of an error generated by supply and substrate coupling, DCDL gain error, DCDL non-linearity, etc., and/or any combination(s) thereof. In some embodiments, the DCDL **116** is implemented by one or more analog and/or digital circuits. For example, the DCDL **116** can be implemented by one or more buffers (e.g., circular buffers) that implement one or more discrete digital logic elements. Alternatively, the DCDL **116** may be implemented by any other analog and/or digital components or elements.

(31) In some embodiments, the error can be reduced and/or mitigated by calibrating one or more components of the PLL **100**, such as the DCDL **116**, using at least a first error injection circuit **118** (identified by FB ERROR INJECTION) and a second error injection circuit **120** (identified by REF ERROR INJECTION), each of which can be controlled at least in part by digital logic **122**. The first error injection circuit **118** of this example has an input (e.g., an error injection input) coupled to an output (e.g., a delay line output, a DCDL output) of the DCDL **116**. The first error injection circuit **118** has an output (e.g., an error injection output) coupled to an input of the PD **106**.

(32) The second error injection circuit **120** of this example has an input (e.g., an error injection input) configured to receive the reference clock signal **102**. The second error injection circuit **120** has an output (e.g., an error injection output) coupled to an input of the PD **106**. Alternatively, the first error injection circuit **118** and/or the second error injection circuit **120** may be disposed elsewhere in the PLL **100**. For example, the first error injection circuit **118** and/or the second error

injection circuit **120** may be coupled between an input of the MMD **114** and an input of the DCDL **116** (identified by ALT ERROR INJECTION **124**). In some embodiments, the first error injection circuit **118** and/or the second error injection circuit **120**, or portion(s) thereof, can be in circuit elsewhere along a reference path of the PLL **100**, which can be a path that includes at least one of the PD **106**, the LF **110**, or the VCO **112**. In some embodiments, the first error injection circuit **118** and/or the second error injection circuit **120**, or portion(s) thereof, can be in circuit elsewhere along a feedback path of the PLL **100**, which can be a path that includes at least one of the VCO **112**, the MMD **114**, the DCDL **116**, or the PD **106**.

(33) In some embodiments, the MMD **114**, the DCDL **116**, the first error injection circuit **118**, and/or, more generally, the PLL **100**, can be configured to cause generation of the feedback clock signal **108** to reduce the error detected by the PD **106**. For example, the digital logic **122** includes control circuitry **126**, which can be implemented by one or more control circuits, to configure at least one of the MMD **114** or the DCDL **116** to reduce a difference between the reference clock signal **102** and the feedback clock signal **108**. In some embodiments, the reduction of the difference can be implemented by shifting the feedback clock signal **108** to have the same error as the reference clock signal **102**.

(34) In the illustrated example, the control circuitry **126** includes a fractional-N (Frac-N) controller **128** to implement Frac-N control logic. For example, the Frac-N controller **128** can change (e.g., periodically change) the division ratio of the MMD **114** from N to $N+1$ and back such that the average is $N+F/M$ where $0 \leq F < M$ and N , F , and M are integers. The control circuitry **126** of this example includes a phase controller **130** to implement digitally controlled delay (DCD) calibration control logic.

(35) The control circuitry **126** of this example can receive first control signal(s) **132** (identified by Frac-N Control), such as digital code(s) (e.g., digital code word(s)), to set an initial configuration of the control circuitry **126**, or portion(s) thereof (e.g., the Frac-N controller **128**). The control circuitry **126** of this example can receive second control signals **134**, **136**. The second control signals **134**, **136** are pseudorandom binary sequence (PRBS) signals respectively generated by PRBS generators **138**, **140**. For example, a first PRBS generator **138** (identified by PRBS-1 GEN) can generate and/or output first PRBS signals **134** (identified by PRBS-1) and a second PRBS generator **140** (identified by PRBS-2 GEN) can generate and/or output second PRBS signals **136** (identified by PRBS-2).

(36) In some embodiments, the PRBS generators **138**, **140** can be implemented by one or more analog circuits and/or one or more digital circuits configured to generate signals based on pseudorandom binary sequences (also referred to as pseudorandom binary codes or pseudorandom bitstreams). Pseudorandom binary sequences are binary sequences that are generated by a deterministic algorithm but are difficult to predict and can exhibit statistical behavior similar to a truly random sequence.

(37) In some embodiments, the digital logic **122** is configured to receive the error signal **107** from the PD **106**. For example, the digital logic **122** can determine whether the error signal **107** is greater than or less than a voltage threshold (e.g., 0 volts (V), 0.5 V, etc.) for each clock cycle of the PD **106**. In some embodiments, the digital logic **122** can generate a first digital code based on the error signal **107** and output the first digital code to the MMD **114** to change a configuration of the MMD **114**. The configuration of the MMD **114** can cause a change in a time delay that the MMD **114** applies to the output clock signal **104**. In some embodiments, the digital logic **122** can generate a second digital code based on the error signal **107** and output the second digital code to the DCDL **116** to change a configuration of the DCDL **116**. The configuration of the DCDL **116** can cause a change in a time delay that the DCDL **116** applies to the delayed clock signal from the MMD **114**.

(38) In the illustrated example, the control circuitry **126**, and/or, more generally, the digital logic **122**, has a first input (e.g., a first control input, a first digital logic input) coupled to an output of the PD **106** and configured to receive the error signal **107**. In this example, the control circuitry **126**,

and/or, more generally, the digital logic **122**, has a second input (e.g., a second control input, a second digital logic input) coupled to an output of the DCDL **116** and configured to receive the delayed clock signal from the DCDL **116**.

(39) In the illustrated example, a first output (e.g., a first control output) of the control circuitry **126**, and/or, more generally, the digital logic **122**, is coupled to an input of the MMD **114**. In this example, a second output (e.g., a second control output) of the control circuitry **126**, and/or, more generally, the digital logic **122**, is coupled to an input of the DCDL **116**. In this example, a third output (e.g., a third control output) of the control circuitry **126**, and/or, more generally, the digital logic **122**, is coupled to an input of the first error injection circuit **118**. In this example, a fourth output (e.g., a fourth control output) of the control circuitry **126**, and/or, more generally, the digital logic **122**, is coupled to an input of the second error injection circuit **120**.

(40) In example operation, the MMD **114** can receive the output clock signal **104**. The MMD **114** can divide the output clock signal by a divisor to generate a divided clock signal. The MMD **114** can delay the divided clock signal by a first time delay to generate a first delayed clock signal. Alternatively, the MMD **114** can pass the output clock signal **104** or the divided clock signal without delay to the DCDL **116**.

(41) In example operation, the DCDL **116** can receive a clock signal. For example, the DCDL **116** can receive the output clock signal **104** to be output from the VCO **112** through the MMD **114**. For example, the MMD **114** can pass the output clock signal **104** without delay from the VCO **112** to the DCDL **116**. In some embodiments, the DCDL **116** can receive the first delayed signal from the MMD **114**. The DCDL **116** can delay the clock signal (e.g., the output clock signal **104**, the first delayed clock signal) by a second time delay to generate a second delayed clock signal.

(42) In example operation, the phase controller **130** can instruct, in response to a first one of the second PRBS signals **136**, the first error injection circuit **118** to delay the second delayed clock signal by a third time delay. Alternatively, the phase controller **130** can instruct, in response to a second one of the second PRBS signals **136**, the first error injection circuit **118** to pass the second delayed clock signal without delay to the PD **106**.

(43) In example operation, the phase controller **130** can instruct, in response to a first one of the first PRBS signals **134**, the second error injection circuit **120** to delay the reference clock signal **102** by a fourth time delay. Alternatively, the phase controller **130** can instruct, in response to a second one of the first PRBS signals **134**, the second error injection circuit **120** to pass the reference clock signal **102** without delay to the PD **106**.

(44) Beneficially, the phase controller **130** can determine to direct at least one of the first error injection circuit **118** or the second error injection circuit **120** to delay a received clock signal such that a phase difference detected by the PD **106** is reduced. Beneficially, in response to at least one of the first error injection circuit **118** or the second error injection circuit **120** introducing and/or injecting an error signal into the PLL **100**, the phase controller **130**, and/or, more generally, the control circuitry **126**, and/or, more generally, the digital logic **122**, can calibrate the DCDL **116** to reduce DCDL gain error. For example, by introducing and/or injecting known errors (e.g., errors from the first and/or second error injection circuits **118**, **120**), the digital logic **122** can cause calibration of the DCDL **116** to reduce DCDL gain error.

(45) While an example implementation of the PLL **100** is depicted in FIG. **1**, other implementations are contemplated. For example, one or more blocks, components, functions, etc., of the PLL **100** may be combined or divided in any other way. The PLL **100** of the illustrated example may be implemented by hardware alone, or by a combination of hardware, software, and/or firmware. For example, the PLL **100** may be implemented by one or more analog circuits (e.g., capacitors, comparators, diodes, inductors, operational amplifiers, resistors, transistors, etc.), one or more digital circuits (e.g., logic gates, etc.), one or more hardware-implemented state machines, one or more programmable processors, one or more application specific integrated circuits (ASICs), etc., and/or any combination(s) thereof. The PLL **100** of the illustrated example can be implemented by

one or more integrated circuits (ICs) on the same die or two or more different dies.

(46) FIG. 2 is a plot **200** depicting example operation of the error injection circuits **118**, **120** of FIG. 1. The plot **200** has an x-axis **202** representing a range of phase codes, which can be digital codes that correspond to a desired and/or expected phase of a PLL, such as the PLL **100** of FIG. 1. The plot **200** has a y-axis **204** representing a range of time delays (identified by $t_{sub.d}$). The plot **200** of the illustrated example can represent a gain of a DCDL, such as the DCDL **116** of FIG. 1. For example, the plot **200** can represent a time delay that may be needed to achieve a desired and/or expected phase of the PLL **100** in accordance with an ideal transfer function for the DCDL.

(47) In the illustrated example, the range of phase codes can include a first range **206** and a second range **208**. The first range **206** can correspond to time delays that can be achieved using the DCDL **116** alone, or by using at least one of the MMD **114** or the DCDL **116** (e.g., a combination of the MMD **114** and/or the DCDL **116**). For example, the DCDL **116** can be used to delay a signal, such as a clock signal, up to at least a first time delay identified by $t_{sub.d1}$. In some embodiments, the first time delay can correspond to a period of the VCO **112** of FIG. 1 ($T_{sub.VCO}$). The second range **208** can correspond to time delays that can be achieved using a combination of the MMD **114** and the DCDL **116**. For example, the second range **208** can begin at the beginning of the first range **206** or at the end of the first range **206**.

(48) In some embodiments, a combination of the MMD **114** and the DCDL **116** can be used to delay a signal, such as a clock signal, up to at least a second time delay identified by $t_{sub.d2}$, which is greater than $t_{sub.d1}$. In some embodiments, the second time delay can correspond to twice of a period of the VCO **112** of FIG. 1 ($2 \cdot T_{sub.VCO}$). Beneficially, in some embodiments, by using the MMD **114** and the DCDL **116**, the time delay range can be extended from $t_{sub.d1}$ to $t_{sub.d2}$.

(49) The example plot **200** depicts an ideal transfer function **210**. The ideal transfer function **210** is characterized and/or defined by a gain of 1. Alternatively, the ideal transfer function **210** may be characterized and/or defined by any other gain. In this example, the plot **200** depicts various lock points of the PLL **100**. For example, a first lock point **212** (identified by Point A Lock Point) is depicted and can correspond to a time delay to be applied to the reference clock signal **102** of FIG. 1 such that the PLL **100** can lock without additional delay. A second lock point **214** is depicted and can correspond to a time delay injected into the PLL **100** by an error injection circuit, such as the first error injection circuit **118** of FIG. 1. For example, the first error injection circuit **118** can inject a time delay corresponding to E_0 of the plot **200**. In some embodiments, E_0 can be a first error representing a first time delay that can be implemented by the first error injection circuit **118**. A third lock point **216** is depicted and can correspond to a time delay injected into the PLL **100** by another error injection circuit, such as the second error injection circuit **120** of FIG. 1. For example, the second error injection circuit **120** can inject a time delay corresponding to E_1 of the plot **200**. In some embodiments, E_1 can be a second error representing a second time delay that can be implemented by the second error injection circuit **120**.

(50) In the illustrated example, the plot **200** depicts deviations from the ideal transfer function **210** as the phase codes and corresponding time delays increase. For example, the plot **200** depicts a first deviation **218** and a second deviation **220** from the ideal transfer function **210**. For example, the first deviation **218** can correspond to a first gain of the DCDL **116** being less than $T_{sub.VCO}$ of the VCO **112**. In some such embodiments, the first gain can represent that too little or not enough time delay has been injected into the PLL **100**. In the illustrated example, the second deviation **220** can correspond to a second gain of the DCDL **116** being greater than $T_{sub.VCO}$ of the VCO **112**. In some such embodiments, the second gain can represent that too much time delay has been injected into the PLL **100**.

(51) In the example plot **200**, a combination (e.g., a sum) of E_0 and E_1 (identified by $E_0 + E_1$) can exercise the error of the PLL **100** from a first error region, which may be defined by the first range **206**, to a second error region, which may be defined by the second range **208** (e.g., a portion of the

second range **208** that extends beyond the first range **206**). In some embodiments, the errors (e.g., E0, E1, E0+E1) can be cycled through in any order by way of an independent template (e.g., a PRBS schema). For example, the phase controller **130** can determine, in response to one(s) of the first PRBS signals **134** and/or the second PRBS signals **136** of FIG. **1**, can cause injection of a first delay (e.g., E0), a second delay (e.g., E1), and/or a third delay (e.g., E0+E1) into the PLL **100**. Beneficially, by scrambling the phase codes through such a PRBS schema, the error can be exercised between the error regions. For example, by injecting delay based on randomly generating PRBS signals, the average injected delay over time may be discernable as illustrated by the first and second deviations **218**, **220**. In some embodiments, the phase controller **130**, and/or, more generally, the digital logic **122** can detect the deviations **218**, **220** from the ideal transfer function **210** and, thusly, determine a subsequent injection of error to compensate and/or mitigate the detected error of the PLL **100**.

(52) FIG. **3** depicts an example implementation of a first error injection circuit **302** and a second error injection circuit **304**. In some embodiments, FIG. **3** can represent controlled error addition such that an input clock signal **306** (identified by CLK) can be delayed and output as an output clock signal **308** (identified by CLKOUT) based on a combination of the first error injection circuit **302** and the second error injection circuit **304**.

(53) In some embodiments, the first error injection circuit **302** can implement the first error injection circuit **118** of FIG. **1**. For example, the first error injection circuit **302** can delay a signal by E0 of FIG. **2**. In some embodiments, the second error injection circuit **304** can implement the second error injection circuit **120** of FIG. **2**. For example, the second error injection circuit **304** can delay a signal by E1 of FIG. **2**. In some embodiments, the combination of the first error injection circuit **302** and the second error injection circuit **304** can delay a signal by E0+E1 of FIG. **2**.

(54) The first error injection circuit **302** includes a first logic gate **310** and a second logic gate **312**. The first logic gate **310** of this example is a NAND logic gate and the second logic gate is an inverter logic gate. Alternatively, the first error injection circuit **302** may be implemented using any other combination of logic gates than those depicted in the example of FIG. **3**. In FIG. **3**, the input clock signal **306** is provided to a first input (e.g., a first multiplexer input) (identified by A) of a first multiplexer **314**. An output of the first logic gate **310** is coupled to an input of the second logic gate **312**. For example, the first logic gate **310** can be coupled to the second logic gate **312** via one or more electrical connections as described herein. An output of the second logic gate **312** is coupled to a second input (e.g., a second multiplexer input) (identified by B) of the multiplexer **314**.

(55) The first logic gate **310** is configured to receive the input clock signal **306** and an enable signal **316** (identified by EN). In some embodiments, the input clock signal **306** can correspond to the output signal from the DCDL **116** of FIG. **1**. The first logic gate **310** can output a logic low signal (e.g., a signal representative of a digital zero, a logic zero) to the second logic gate **312** in response to the clock signal **306** and the enable signal **316** being logic high signals. The second logic gate **312** can invert the logic low signal to a logic high signal (e.g., a signal representative of a digital one, a logic one) and output the logic high signal to the first multiplexer **314**.

(56) In some embodiments, the first error injection circuit **302** can pass, provide, and/or output the input clock signal **306** without delay to the first multiplexer **314**. For example, the phase controller **130**, and/or, more generally, the digital logic **122** of FIG. **1**, can generate a first selection signal **318** (identified by MUXSELBUF<1>) to select the first input such as to output the input clock signal **306** to the second error injection circuit **304** without delay. In some embodiments, the first selection signal **318** can be generated in response to one of the first PRBS signals **134**.

(57) In some embodiments, the first error injection circuit **302** can delay the input clock signal **306** to generate a first delayed clock signal **320** (identified by CLKDEL0). For example, the phase controller **130**, and/or, more generally, the digital logic **122** of FIG. **1**, can generate the first selection signal **318** to select the second input such as to output the first delayed clock signal **320** to

the second error injection circuit **304**. In some embodiments, the first selection signal **318** can be generated in response to another one of the first PRBS signals **134**.

(58) The second error injection circuit **304** includes a third logic gate **322** and a fourth logic gate **324**. The third logic gate **322** and the fourth logic gate **324** of this example are inverter logic gates. Alternatively, the second error injection circuit **304** may be implemented using any other combination of logic gates than those depicted in the example of FIG. 3. In FIG. 3, the output signal from the first multiplexer **314** is provided to a first input (e.g., a first multiplexer input) (identified by A) of a second multiplexer **326**. An output of the third logic gate **322** is coupled to an input of the fourth logic gate **324**. For example, the third logic gate **322** can be coupled to the fourth logic gate **324** via one or more electrical connections as described herein. An output of the fourth logic gate **324** is coupled to a second input (e.g., a second multiplexer input) (identified by B) of the second multiplexer **326**.

(59) The third logic gate **322** is configured to receive the output signal from the first multiplexer **314**. For example, the third logic gate **322** can receive a logic high signal, invert the logic high signal to a logic low signal, and output the logic low signal to the fourth logic gate **324**. The fourth logic gate **324** can invert the logic low signal to a logic high signal and output the logic high signal to the second multiplexer **326** as a second delayed clock signal **328** (identified by CLKDEL1). For example, the path along the third and fourth logic gates **322**, **324** can delay the output signal from the first multiplexer **314** by a time delay that may correspond to E1 of FIG. 2.

(60) In some embodiments, the second error injection circuit **304** can pass, provide, and/or output the output signal from the first multiplexer **314** without delay to the second multiplexer **326**. For example, the phase controller **130**, and/or, more generally, the digital logic **122** of FIG. 1, can generate a second selection signal **330** (identified by MUXSELBUF<0>) to select the first input such as to output the output signal as the output clock signal **308** without delay. In some embodiments, the second selection signal **330** can be generated in response to one of the second PRBS signals **136**. In some embodiments, the output clock signal **308** can correspond to the delayed clock signal output from the second error injection circuit **120** of FIG. 1.

(61) In some embodiments, the second error injection circuit **304** can delay the output of the first multiplexer **314**, which can be the input clock signal **306** or the first delayed clock signal **320**, to generate the second delayed clock signal **328**. For example, the phase controller **130**, and/or, more generally, the digital logic **122** of FIG. 1, can generate the second selection signal **330** to select the second input such as to output the second delayed clock signal **328** as the output clock signal **308**. In some embodiments, the second selection signal **330** can be generated in response to another one of the second PRBS signals **136**.

(62) In some embodiments, the selection signals **318**, **330** can be buffered. For example, multiplexer selection signal(s) **332** can be provided to a fifth logic gate **334**, which is coupled to a sixth logic gate **336**, both of which are inverter logic gates. The sixth logic gate **336** can output the selection signals **318**, **330**. Alternatively, the multiplexer selection signal(s) **332** can be buffered using any other combination of logic gates.

(63) FIG. 4 depicts another example implementation of a first error injection circuit **402** (identified by ERROR E0) and a second error injection circuit **404** (identified by ERROR E1). In some embodiments, FIG. 4 can represent controlled error addition such that an input clock signal **406** (identified by CLK) can be delayed and output as an output clock signal **408** (identified by CLKOUT) based on a combination of the first error injection circuit **402** and the second error injection circuit **404**. In some embodiments, the second error injection circuit **404** can change a time delay at which it can delay a signal and/or otherwise is adjustable.

(64) In some embodiments, the first error injection circuit **402** can implement the first error injection circuit **118** of FIG. 1. For example, the first error injection circuit **402** can delay a signal by E0 of FIG. 2. In some embodiments, the second error injection circuit **404** can implement the second error injection circuit **120** of FIG. 2. For example, the second error injection circuit **404** can

delay a signal by E1 of FIG. 2 or, in some embodiments, E1a or E1b as described below. In some embodiments, the combination of the first error injection circuit **402** and the second error injection circuit **404** can delay a signal by E0+E1 of FIG. 2 or, in some embodiments, E0+E1a or E0+E1b as described below.

(65) In the illustrated example, a D-flip-flop **410** (identified by DFF) is configured to receive the input clock signal **406** at an input (identified by D) The D-flip-flop **410** can be triggered in response to a clock edge (e.g., a rising edge, a falling edge) of a reference clock signal **412** (identified by RCLK). The D-flip-flop **410** can be configured to output the input clock signal **406** to the first error injection circuit **402**. Alternatively, the D-flip-flop **410** may be any other type of flip-flop.

(66) The first error injection circuit **402** of this example is implemented by a latch **414**. Alternatively, the first error injection circuit **402** may be implemented by the D-flip-flop **410** and the latch **414**. The latch **414** can be configured to delay the input clock signal **406** by a time delay to generate a first delayed clock signal **416** (identified by DCLK0). The latch **414** can be triggered in response to an edge (e.g., a rising edge, a falling edge) of the reference clock signal **412**. The output of the latch **414** is coupled to a first input (identified by A) of a first multiplexer **418**.

(67) In the illustrated example, the first error injection circuit **402** can be bypassed via a first logic gate **420** and a second logic gate **422**. The first and second logic gates **420**, **422** of this example are inverter logic gates. An output (identified by Q) of the D-flip-flop **410** is coupled to an input of a first logic gate **420**. An output of the first logic gate **420** is coupled to an input of a second logic gate **422**. An output of the second logic gate **422** is coupled to a second input (identified by B) of the first multiplexer **418**.

(68) In some embodiments, the phase controller **130**, and/or, more generally, the digital logic **122** of FIG. 1, can generate a first multiplexer selection signal **424** (identified by MUXSEL<1>) to select the first input such as to output the first delayed clock signal **416** as an output of the first multiplexer **418**. In some embodiments, the first multiplexer selection signal **424** can be generated in response to one of the first PRBS signals **134**.

(69) In some embodiments, the phase controller **130**, and/or, more generally, the digital logic **122**, can generate the first multiplexer selection signal **424** such as to pass the input clock signal **406** to a third logic gate **426** without delay from the first error injection circuit **402** and thereby bypass the first error injection circuit **402**. In some embodiments, the first multiplexer selection signal **424** can be generated in response to a different one of the first PRBS signals **134**.

(70) The third logic gate **426** is an inverter logic gate and has an output coupled to a fourth logic gate **428** and a fifth logic gate **430** via a node **432**. The third logic gate **426** and the fourth logic gate **428** of this example are inverter logic gates. The fifth logic gate **430** is a NAND logic gate. A first input (identified by A) of the fifth logic gate **430** is coupled to a sixth logic gate **434**, which is an inverter logic gate. A second input (identified by B) of the fifth logic gate **430** is coupled to an output of the third logic gate **426**.

(71) An output of the fifth logic gate **430** is coupled to the second error injection circuit **404**, which has an output coupled to a first input (identified by A) of a second multiplexer **436**. An output of the fourth logic gate **428** is coupled to a second input (identified by B) of the second multiplexer **436**. An output of a sixth logic gate **438**, which is an inverter logic gate, is coupled to a third input (identified by C) of the second multiplexer **436**. A fourth input (identified by D) of the second multiplexer **436** is configured to receive a signal **439** (identified by LVR_PD). In some embodiments, the signal **439** is a low-voltage release (LVR) signal from the PD **106** of FIG. 1. Alternatively, the second multiplexer **436** may not have the fourth input or may be configured to receive a different signal. An output of the second multiplexer **436** is coupled to a seventh logic gate **440**, which has an output coupled to an input of a buffer **442**. The seventh logic gate **440** of this example is an inverter logic gate.

(72) In some embodiments, the second error injection circuit **404** is implemented by a DCDL **444**.

For example, the DCDL can include a plurality of buffers that can be individually configured to be in a delay chain or isolated from the delay chain. In some such embodiments, the DCDL can enable one or more buffers to be part of the delay chain and/or disable one or more buffers from being part of the delay chain.

(73) In some embodiments, the first multiplexer **418** can output the input clock signal **406** (e.g., output without delay E0) or the first delayed clock signal **416** to the third logic gate **426**. The third logic gate **426** can invert the output to generate a first clock signal **446** (identified by CLK1). The third logic gate **426** can provide the first clock signal **446** to the fifth logic gate **430** and input B via the fourth logic gate **428**.

(74) In some embodiments, the second multiplexer **436** can output the input clock signal **406** without delay or the input clock signal **406** delayed by at least one of the first error injection circuit **402** or the second error injection circuit **404**. For example, in response to a first value of a second multiplexer selection signal **448** (identified by SMUXSEL<3:0>), the second multiplexer **436** can output a signal received by input A, such as a second delayed clock signal **450** (identified by DCLK1). In some embodiments, the second delayed clock signal **450** can correspond to the input clock signal **406** delayed by the first error injection circuit **402** and/or the second error injection circuit **404**. In some embodiments, in response to a second value of the second multiplexer selection signal **448**, the second multiplexer **436** can output a signal received by input B, such as an inversion of the first clock signal **446**. In some embodiments, the first clock signal **446** can correspond to the input clock signal **406** without delay or the input clock signal **406** delayed by the first error injection circuit **402** (and thereby bypassing the second error injection circuit **404**). In some embodiments, in response to a third value of the second multiplexer selection signal **448**, the second multiplexer **436** can output a signal received by input C, such as an inversion of a bypass signal **452**. In some embodiments, the bypass signal **452** can be generated after a determination, such as by the phase controller **130**, that the input clock signal **406** is to be passed through the circuit without delay. For example, the bypass signal **452** can be selected for output by the second multiplexer **436** to bypass the first error injection circuit **402** and the second error injection circuit **404**.

(75) In some embodiments, the fifth logic gate **430** can receive the first clock signal **446** via the third logic gate **426**. The first clock signal **446** can be either the first delayed clock signal **416**, which is the input clock signal **406** delayed by E0, or the input clock signal **406** without delay. If the first clock signal **446** is a logic high signal and the output of the sixth logic gate **434** is a logic low signal (e.g., signal SEL<1> is a logic high signal and inverted by the sixth logic gate **434**), the fifth logic gate **430** can enable the DCDL **444** to delay an input of the DCDL **444** by a time delay. In some embodiments, the time delay is adjustable in response to an error control signal **454** (identified by E1CTRL<N:0>). The error control signal **454** can be a digital code in a range of 0 to N. In some embodiments, the error control signal **454** can instruct the DCDL **444** to implement a first time delay of E1a or a second time delay of E1b. For example, the second error injection circuit **404** can delay the first clock signal **446** by E1a or E1b (or any other number of values, e.g., E1c, E1d, etc.) to generate the second delayed clock signal **450**. In some embodiments, in response to the first value of the second multiplexer selection signal **448**, the second multiplexer **436** can output the second delayed clock signal **450** as the output clock signal **408**. In some embodiments, the error control signal **454** can alternate such that during one clock cycle E1a can be used and during a subsequent clock cycle E1b can be used before returning to using E1a. Alternatively, the error control signal **454** may be generated such that E1a and E1b are chosen in any type of order.

(76) In some embodiments, one or more of the signals depicted in the example of FIG. 4 can be generated by the phase controller **130**, and/or, more generally, the control circuitry **126**, and/or, more generally, the digital logic **122** of FIG. 1. For example, the phase controller **130** can generate at least one of the reference clock signal **412**, value(s) of the first multiplexer selection signal **424**, value(s) of the second multiplexer selection signal **448**, the bypass signal **452**, value(s) of the error

control signal **454**, or a selection signal **456** (identified by SEL<1>).

(77) FIG. 5 is a plot **500** of an example calibration operation of the digitally controlled delay line of FIG. 1 in response to a detection of negative error. The plot **500** has an x-axis **502** representing a range of phase codes, which can be digital codes that correspond to a desired and/or expected phase of a PLL, such as the PLL **100** of FIG. 1. The plot **500** has a y-axis **504** representing a range of time delays (identified by t.sub.d). The plot **500** of the illustrated example can represent a gain of a DCDL, such as the DCDL **116** of FIG. 1. For example, the plot **500** can represent a time delay that may be needed to achieve a desired and/or expected phase of the PLL **100** in accordance with an ideal transfer function **506** for the DCDL.

(78) The ideal transfer function **506** of this example is characterized and/or defined by a gain of 1. Alternatively, the ideal transfer function **506** may be characterized and/or defined by any other gain. In this example, the plot **500** depicts various lock points of the PLL **100**. For example, a first lock point **508** (identified by Point A Lock Point) is depicted and can correspond to a time delay to be applied to the reference clock signal **102** of FIG. 1 such that the PLL **100** can lock without additional delay.

(79) A second lock point **510** is depicted and can correspond to a time delay injected into the PLL **100** by an error injection circuit, such as the first error injection circuit **118** of FIG. 1. For example, the first error injection circuit **118** of FIG. 1, the first error injection circuit **302** of FIG. 3, and/or the first error injection circuit **402** of FIG. 4 can inject a time delay corresponding to E0 of the plot **500**. In some embodiments, E0 can be a first error representing a first time delay that can be implemented by the first error injection circuit **118** of FIG. 1, the first error injection circuit **302** of FIG. 3, and/or the first error injection circuit **402** of FIG. 4.

(80) A third lock point **512** is depicted and can correspond to a time delay injected into the PLL **100** by another error injection circuit, such as the second error injection circuit **120** of FIG. 1, the second error injection circuit **304** of FIG. 3, and/or the second error injection circuit **404** of FIG. 4. For example, the second error injection circuit **120** of FIG. 1, the second error injection circuit **304** of FIG. 3, and/or the second error injection circuit **404** of FIG. 4 can inject a time delay corresponding to E1 of the plot **200**. In some embodiments, E1 can be a second error representing a second time delay that can be implemented by the second error injection circuit **120** of FIG. 1, the second error injection circuit **304** of FIG. 3, and/or the second error injection circuit **404** of FIG. 4.

(81) A fourth lock point **514** is depicted and can correspond to a time delay injected into the PLL **100** by a combination of error injection circuits, such as a combination of the first and second error injection circuits **118**, **120** of FIG. 1, the first and second error injection circuits **302**, **304** of FIG. 3, and/or the first and second error injection circuits **402**, **404** of FIG. 4. For example, the first and second error injection circuits **118**, **120** of FIG. 1 can inject a time delay corresponding to a combination of E0 and E1 of the plot **500**. In some embodiments, E0+E1 can be a third error representing a second time delay that can be implemented by a combination of error injection circuits.

(82) In the illustrated example, the plot **500** depicts example calibration in response to the error injections. For example, the Frac-N controller **128** and/or the phase controller **130** can calibrate at least one of the MMD **114** or the DCDL **116** to reduce a gain error. For example, the digital logic **122** can calibrate the DCDL **116** to reduce a gain error of the DCDL **116** after injection(s) of E0, E1, or E0+E1 into the PLL **100**.

(83) By way of example, in response to an injection of E0 into the PLL **100**, the digital logic **122** can detect a first error **516** (e.g., an error detected by the PD **106**) (identified by $\delta_{\text{sub.0}}$), which can represent a difference between an expected time delay of the ideal transfer function **506** at a phase code and an observed second time delay at the phase code. Beneficially, the digital logic **122** can calibrate the DCDL **116** to reduce the first error **516** and, thusly, reduce the gain error of the DCDL **116**. In the example plot **500**, the calibration can correspond to an amount of time delay **518** (identified by E.sub.0|cal). The time delay amount **518** can be determined based on a difference

between E0 (e.g., the time delay defined by and/or corresponding to E0) and the first error **516**.
(84) By way of another example, in response to an injection of E1 into the PLL **100**, the digital logic **122** can detect a second error **520** (identified by δ .sub.1), which can represent a difference between an expected time delay of the ideal transfer function **506** at a phase code and an observed second time delay at the phase code. Beneficially, the digital logic **122** can calibrate the DCDL **116** to reduce the second error **520** and, thusly, reduce the gain error of the DCDL **116**. In the example plot **500**, the calibration can correspond to an amount of time delay **522** (identified by E.sub.1|cal). The time delay amount **522** can be determined based on a difference between E1 (e.g., the time delay defined by and/or corresponding to E1) and the second error **520**.

(85) In the illustrated example, the time delays represented by E0, E1, and E1+E2 cross into a second region **524** (identified by REGION II) of the plot **500**. Beneficially, the time delays have values such that the detected errors can be pushed from a first region **526** (identified by REGION I) into the second region **524** for improved error detection. For example, if E0, E1, and/or E1+E2 caused the error to be in the first region **526**, the error may be indistinguishable from other error sources.

(86) FIG. **6** depicts an example dead-zone analysis **600** for determination of error values. In the illustrated example, a period (T.sub.VCO) of the VCO **112** of FIG. **1** is 70 picoseconds (ps) and half the period is 35 ps. In the illustrated example, a first time delay that can be injected and/or implemented by a first error injection circuit, such as the first error injection circuit **118** of FIG. **1**, is E0 with a value of 37 ps. In the illustrated example, a second time delay that can be injected and/or implemented by a second error injection circuit, such as the second error injection circuit **120** of FIG. **1**, is E1 with a value of 40 ps.

(87) In FIG. **6**, a first region **602** (identified by RI) is defined from 0 to half T.sub.VCO, which is 35 ps. A second region **604** (identified by RII) is defined from half T.sub.VCO to T.sub.VCO, which is 70 ps. For example, the first region **602** of FIG. **6** can correspond to the first region **526** of FIG. **5**. In some embodiments, the second region **604** of FIG. **6** can correspond to the second region **524** of FIG. **5**.

(88) In some embodiments, a phase code that corresponds to a phase of the PLL **100** in a phase range of 0-30 ps and an error injection of E0 or E1 can result in a gain of the DCDL **116** of FIG. **1** to be in the first region **602**. A phase code that corresponds to a phase of the PLL **100** in a phase range of 0-30 ps and an error injection of E0 and E1 (E0+E1) can result in a gain of the DCDL **116** of FIG. **1** to be in the second region **604**. In some such embodiments, the detected error can be positive (+ve).

(89) In some embodiments, a phase code that corresponds to a phase of the PLL **100** in a phase range of 30-33 ps and an error injection of E0 can result in a gain of the DCDL **116** of FIG. **1** to be in the first region **602**. A phase code that corresponds to a phase of the PLL **100** in a phase range of 30-33 ps and an error injection of E1 can result in a gain of the DCDL **116** of FIG. **1** to be in the second region **604**. In some such embodiments, the detected error is indeterminate because of the mismatch between E0 and E1. Thusly, the phase range of 30-33 ps, based on the example values of E0 and E1, may create a first error detection dead-zone **606** (identified by DEAD-ZONE 1) in this phase range. Similarly, a second error detection dead-zone **608** (identified by DEAD-ZONE 2) may exist in the phase range of 63-70 ps due to the combination of E0 and E1 being greater than T.sub.VCO. In some embodiments disclosed herein, the values of E0 and E1 can be selected such that the error detection dead-zones **606**, **608** may be avoided.

(90) FIG. **7** depicts another example dead-zone analysis **700** for determination of error values. In the illustrated example, a period (T.sub.VCO) of the VCO **112** of FIG. **1** is 70 ps and half the period is 35 ps. In the illustrated example, a first time delay that can be injected and/or implemented by a first error injection circuit, such as the first error injection circuit **118** of FIG. **1**, is E0 with a value of 37 ps. In the illustrated example, a second time delay that can be injected and/or implemented by a second error injection circuit, such as the second error injection circuit **120** of FIG. **1**, is E1 with a

value of 29 ps.

(91) In FIG. 7, a first region **702** (identified by RI) is defined from 0 to half T.sub.VCO, which is 35 ps. A second region **704** (identified by RII) is defined from half T.sub.VCO to T.sub.VCO, which is 70 ps. For example, the first region **702** of FIG. 7 can correspond to the first region **526** of FIG. 5. In some embodiments, the second region **704** of FIG. 7 can correspond to the second region **524** of FIG. 5.

(92) In the illustrated example, a third error detection dead-zone **706** (identified by DEAD-ZONE 3) may exist in the phase range of 0-4 ps due to the combination of E0 and E1 being less than T.sub.VCO. In the illustrated example, a fourth error detection dead-zone **708** (identified by DEAD-ZONE 4) may exist because of a mismatch between E0 and E1. For example, a phase code that corresponds to a phase of the PLL **100** in a phase range of 33-41 ps and an error injection of E0 can result in a gain of the DCDL **116** of FIG. 1 to be in the first region **702**. A phase code that corresponds to a phase of the PLL **100** in the phase range of 33-41 ps and an error injection of E1 can result in a gain of the DCDL **116** of FIG. 1 to be in the second region **704**. In some such embodiments, the detected error is indeterminate because of the mismatch between E0 and E1. Thusly, the phase range of 33-41 ps, based on the example values of E0 and E1, may create the fourth error detection dead-zone **708** in this phase range. In some embodiments disclosed herein, the values of E0 and E1 can be selected such that the error detection dead-zones **706**, **708** may be avoided.

(93) FIG. 8 depicts a combination **800** of the example dead-zone analyses **600**, **700** of FIGS. 6 and/or 7 for determination of error values. For example, there may be error detection dead-zones **606**, **608**, **706**, **708** in the phase ranges of 0-4, 30-33, 33-41, and 63-70 ps. Beneficially, in some embodiments, error E1 can be adjustable such that it can have different error values to avoid the error detection dead-zones **606**, **608**, **706**, **708**. For example, selecting two values of E1 such that:

$E1a < E0 < E1b$;
 $E1a + E0 < T_{\text{sub.VCO}}$; and
 $E1b + E0 > T_{\text{sub.VCO}}$,

which can enable the avoidance of the error detection dead-zones **606**, **608**, **706**, **708**.

(94) Exemplary conditions that may be satisfied to avoid the error detection dead-zones **606**, **608**, **706**, **708** are as follows:

(95) $E1a = E0 - \epsilon_a = \frac{T_{\text{VCO}}}{2} + \epsilon_0 - \epsilon_a$,

Condition I, where E0 can be

(96) $\frac{T_{\text{VCO}}}{2} + \epsilon_0$

(e.g., E0 is approximately half the period of the VCO **112**);

(97) $E1b = E0 + \epsilon_b = \frac{T_{\text{VCO}}}{2} + \epsilon_0 + \epsilon_b$,

Condition II; and

$E1a + E0 < T_{\text{sub.VCO}}$ Condition III.

(98) Condition III may be evaluated using Conditions I and II such that:

(99) $\frac{T_{\text{VCO}}}{2} + \epsilon_0 - \epsilon_a + \frac{T_{\text{VCO}}}{2} + \epsilon_0 < T_{\text{VCO}}$,

which yields $2\epsilon_{\text{sub.0}} < \delta_{\text{sub.a}}$.

(100) An additional condition may be:

$E1a + E0 < T_{\text{sub.VCO}}$ Condition IV,

which may be evaluated using Conditions I and II such that:

(101) $\frac{T_{\text{VCO}}}{2} + \epsilon_0 - \epsilon_b + \frac{T_{\text{VCO}}}{2} + \epsilon_0 > T_{\text{VCO}}$,

which yields $2\epsilon_{\text{sub.0}} > -\delta_{\text{sub.b}}$.

(102) In the example conditions above, $\epsilon_{\text{sub.0}}$ can represent additional analog delay. For example, $\epsilon_{\text{sub.0}}$ can result from circuit manufacturing processes, thermal characteristics of circuit components, and/or the like. In the example conditions above, $\delta_{\text{sub.a}}$ and $\delta_{\text{sub.b}}$ can represent two different values or, in some embodiments, the same value, which can be used to determine E1a and E1b. For example, E1a and E1b can be determined using the example conditions above and

implemented in hardware by the DCDL **444** of FIG. **4**, and/or, more generally, the second error injection circuit **404** of FIG. **4**. For example, the DCDL **444** can be controlled such that it can delay an input signal by E1a or E1b. Beneficially, the values of E1a and E1b can be determined such that the error detection dead-zones **606**, **608**, **706**, **708** of FIGS. **6**, **7**, and/or **8** can be avoided and, thusly, the gain error of the DCDL **116** of FIG. **1** can be calibrated regardless of the phase of the PLL **100**.

(103) FIG. **9** is a flowchart representative of an example process **900** that may be performed and/or implemented using hardware logic or machine-readable instructions that may be executed by processor circuitry to implement the phase-locked loop **100** of FIG. **1**. The process **900** of FIG. **9** begins at block **902**, at which the PLL **100** receives an input clock signal associated with a phase-locked loop. For example, the second error injection circuit **120** can receive the reference clock signal **102**.

(104) At block **904**, the PLL **100** calibrates an error injection circuit based on half of a period of a voltage-controlled oscillator. For example, the digital logic **122** can calibrate the first error injection circuit **118** such that the first error injection circuit **118** can delay an output of the DCDL **116** by half of T.sub.VCO.

(105) At block **906**, the PLL **100** identifies an error margin associated with the error injection circuit to generate a first error value and a second error value. For example, the digital logic **122** can determine at least one of $\delta_{\text{sub.a}}$ or $\delta_{\text{sub.b}}$ to satisfy the example conditions above for determination of a first error value, such as E1a, and a second error value, such as E1b.

(106) At block **908**, the PLL **100** injects a first error signal corresponding to the first error value into the phase-locked loop. For example, the phase controller **130**, in response to one of the first PRBS signals **134** from the first PRBS generator **138**, can instruct the first error injection circuit **118** to inject an error signal such that an output of the DCDL **116** is delayed by an error, such as E0.

(107) At block **910**, the PLL **100** injects a second error signal corresponding to the second error value into the phase-locked loop. For example, the phase controller **130**, in response to one of the second PRBS signals **136** from the second PRBS generator **140**, can instruct the second error injection circuit **120** to inject an error signal such that the reference clock signal **102** is delayed by an error, such as E1a or E1b.

(108) At block **912**, the PLL **100** calibrates a gain error associated with the phase-locked loop based on at least one of the first error signal or the second error signal. For example, the PD **106** can determine a phase difference between the outputs of the first and second error injection circuits **118**, **120**. In some embodiments, the digital logic **122** can calibrate the DCDL **116** based on the phase difference as described at least in connection with FIG. **5**. Beneficially, the digital logic **122** can calibrate the DCDL **116** such that a gain error of the DCDL **116** is reduced. After calibrating the gain error at block **912**, the process **900** of FIG. **9** concludes.

(109) Techniques operating according to the principles described herein may be implemented in any suitable manner. The processing and decision blocks of the flowcharts above represent steps and acts that may be included in algorithms that carry out these various processes. Algorithms derived from these processes may be implemented as software integrated with and directing the operation of one or more single- or multi-purpose processors, may be implemented as functionally equivalent circuits such as a DSP circuit or an ASIC, or may be implemented in any other suitable manner. It should be appreciated that the flowcharts included herein do not depict the syntax or operation of any particular circuit or of any particular programming language or type of programming language. Rather, the flowcharts illustrate the functional information one skilled in the art may use to fabricate circuits or to implement computer software algorithms to perform the processing of a particular apparatus carrying out the types of techniques described herein. For example, the flowcharts, or portion(s) thereof, may be implemented by hardware alone (e.g., one or more analog or digital circuits, one or more hardware-implemented state machines, etc., and/or any combination(s) thereof) that is configured or structured to carry out the various processes of the

flowcharts. In some examples, the flowcharts, or portion(s) thereof, may be implemented by machine-executable instructions (e.g., machine-readable instructions, computer-readable instructions, computer-executable instructions, etc.) that, when executed by one or more single- or multi-purpose processors, carry out the various processes of the flowcharts. It should also be appreciated that, unless otherwise indicated herein, the particular sequence of steps and/or acts described in each flowchart is merely illustrative of the algorithms that may be implemented and can be varied in implementations and embodiments of the principles described herein.

(110) Accordingly, in some embodiments, the techniques described herein may be embodied in machine-executable instructions implemented as software, including as application software, system software, firmware, middleware, embedded code, or any other suitable type of computer code. Such machine-executable instructions may be generated, written, etc., using any of a number of suitable programming languages and/or programming or scripting tools, and also may be compiled as executable machine language code or intermediate code that is executed on a framework, virtual machine, or container.

(111) When techniques described herein are embodied as machine-executable instructions, these machine-executable instructions may be implemented in any suitable manner, including as a number of functional facilities, each providing one or more operations to complete execution of algorithms operating according to these techniques. A “functional facility,” however instantiated, is a structural component of a computer system that, when integrated with and executed by one or more computers, causes the one or more computers to perform a specific operational role. A functional facility may be a portion of or an entire software element. For example, a functional facility may be implemented as a function of a process, or as a discrete process, or as any other suitable unit of processing. If techniques described herein are implemented as multiple functional facilities, each functional facility may be implemented in its own way; all need not be implemented the same way. Additionally, these functional facilities may be executed in parallel and/or serially, as appropriate, and may pass information between one another using a shared memory on the computer(s) on which they are executing, using a message passing protocol, or in any other suitable way.

(112) Generally, functional facilities include routines, programs, objects, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Typically, the functionality of the functional facilities may be combined or distributed as desired in the systems in which they operate. In some implementations, one or more functional facilities carrying out techniques herein may together form a complete software package. These functional facilities may, in alternative embodiments, be adapted to interact with other, unrelated functional facilities and/or processes, to implement a software program application.

(113) Some exemplary functional facilities have been described herein for carrying out one or more tasks. It should be appreciated, though, that the functional facilities and division of tasks described is merely illustrative of the type of functional facilities that may implement using the exemplary techniques described herein, and that embodiments are not limited to being implemented in any specific number, division, or type of functional facilities. In some implementations, all functionalities may be implemented in a single functional facility. It should also be appreciated that, in some implementations, some of the functional facilities described herein may be implemented together with or separately from others (e.g., as a single unit or separate units), or some of these functional facilities may not be implemented.

(114) Machine-executable instructions implementing the techniques described herein (when implemented as one or more functional facilities or in any other manner) may, in some embodiments, be encoded on one or more computer-readable media, machine-readable media, etc., to provide functionality to the media. Computer-readable media include magnetic media such as a hard disk drive, optical media such as a CD or a DVD, a persistent or non-persistent solid-state memory (e.g., Flash memory, Magnetic RAM, etc.), or any other suitable storage media. Such a

computer-readable medium may be implemented in any suitable manner. As used herein, the terms “computer-readable media” (also called “computer-readable storage media”) and “machine-readable media” (also called “machine-readable storage media”) refer to tangible storage media. Tangible storage media are non-transitory and have at least one physical, structural component. In a “computer-readable medium” and “machine-readable medium” as used herein, at least one physical, structural component has at least one physical property that may be altered in some way during a process of creating the medium with embedded information, a process of recording information thereon, or any other process of encoding the medium with information. For example, a magnetization state of a portion of a physical structure of a computer-readable medium, a machine-readable medium, etc., may be altered during a recording process.

(115) Further, some techniques described above comprise acts of storing information (e.g., data and/or instructions) in certain ways for use by these techniques. In some implementations of these techniques—such as implementations where the techniques are implemented as machine-executable instructions—the information may be encoded on a computer-readable storage media. Where specific structures are described herein as advantageous formats in which to store this information, these structures may be used to impart a physical organization of the information when encoded on the storage medium. These advantageous structures may then provide functionality to the storage medium by affecting operations of one or more processors interacting with the information; for example, by increasing the efficiency of computer operations performed by the processor(s).

(116) In some, but not all, implementations in which the techniques may be embodied as machine-executable instructions, these instructions may be executed on one or more suitable computing device(s) and/or electronic device(s) operating in any suitable computer and/or electronic system, or one or more computing devices (or one or more processors of one or more computing devices) and/or one or more electronic devices (or one or more processors of one or more electronic devices) may be programmed to execute the machine-executable instructions. A computing device, electronic device, or processor (e.g., processor circuitry) may be programmed to execute instructions when the instructions are stored in a manner accessible to the computing device, electronic device, or processor, such as in a data store (e.g., an on-chip cache or instruction register, a computer-readable storage medium and/or a machine-readable storage medium accessible via a bus, a computer-readable storage medium and/or a machine-readable storage medium accessible via one or more networks and accessible by the device/processor, etc.). Functional facilities comprising these machine-executable instructions may be integrated with and direct the operation of a single multi-purpose programmable digital computing device, a coordinated system of two or more multi-purpose computing device sharing processing power and jointly carrying out the techniques described herein, a single computing device or coordinated system of computing device (co-located or geographically distributed) dedicated to executing the techniques described herein, one or more FPGAs for carrying out the techniques described herein, or any other suitable system.

(117) Embodiments have been described where the techniques are implemented in circuitry and/or machine-executable instructions. It should be appreciated that some embodiments may be in the form of a method, of which at least one example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

(118) Various aspects of the embodiments described above may be used alone, in combination, or in a variety of arrangements not specifically discussed in the embodiments described in the foregoing and is therefore not limited in its application to the details and arrangement of components set forth in the foregoing description or illustrated in the drawings. For example, aspects described in one embodiment may be combined in any manner with aspects described in other embodiments.

(119) The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both,” of the elements so conjoined, e.g., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, e.g., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B,” when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

(120) The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

(121) As used herein in the specification and in the claims, the phrase, “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently, “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, at least one, optionally including more than one, B (and optionally including other elements); etc.

(122) Use of ordinal terms such as “first,” “second,” “third,” etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

(123) Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

(124) All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

(125) The word “exemplary” is used herein to mean serving as an example, instance, or illustration. Any embodiment, implementation, process, feature, etc., described herein as exemplary should therefore be understood to be an illustrative example and should not be understood to be a preferred or advantageous example unless otherwise indicated.

(126) Having thus described several aspects of at least one embodiment, it is to be appreciated that various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of this disclosure and are intended to be within the spirit and scope of the principles described herein. Accordingly, the foregoing description and drawings are by way of example only.

Claims

1. An apparatus for calibrating a phase-locked loop, comprising: a digitally controlled delay line (DCDL) with a DCDL output and configured to: receive a clock signal to be output from a voltage-controlled oscillator; and delay the clock signal to generate a first delayed clock signal; an error injection circuit with a first error injection input and a second error injection input, the first error injection input coupled to the DCDL output; and a phase controller with a phase controller output coupled to the second error injection input, the phase controller configured to, in response to a pseudorandom binary sequence signal, instruct the error injection circuit to generate a second delayed clock signal based on a delay of the first delayed clock signal.
2. The apparatus of claim 1, wherein the error injection circuit has an error injection output, and further comprising a phase detector with a phase detector input coupled to the error injection output, and wherein the error injection circuit is configured to output the second delayed clock signal to the phase detector.
3. The apparatus of claim 2, wherein the second delayed clock signal is a second feedback clock signal, and the phase detector is configured to: determine a first phase difference between an input clock signal and a first feedback clock signal at a first time; and determine a second phase difference between the input clock signal and the second feedback clock signal at a second time, after the first time, and the second phase difference is less than the first phase difference.
4. The apparatus of claim 1, wherein the pseudorandom binary sequence signal is a first pseudorandom binary sequence signal, and the phase controller is configured to instruct the error injection circuit to pass the first delayed clock signal to a phase detector in response to a second pseudorandom binary sequence signal.
5. The apparatus of claim 1, wherein the error injection circuit is a first error injection circuit, and further comprising: a phase detector with a phase detector input; and a second error injection circuit with a third error injection input and an error injection output, the third error injection input configured to receive an input clock signal, and the error injection output coupled to the phase detector input.
6. The apparatus of claim 1, wherein the phase controller output is a first phase controller output, the phase controller has a second phase controller output, the error injection circuit is a first error injection circuit, the pseudorandom binary sequence signal is a first pseudorandom binary sequence signal, and further comprising: a second error injection circuit with a third error injection input coupled to the second phase controller output and configured to receive an input clock signal, and wherein the phase controller is configured to, in response to a second pseudorandom binary sequence signal, instruct the second error injection circuit to generate a delayed input clock signal based on a delay of the input clock signal.
7. The apparatus of claim 6, wherein the second error injection circuit has an error injection output, and further comprising a phase detector with a phase detector input coupled to the error injection output, and wherein the second error injection circuit is configured to output the delayed input clock signal to the phase detector.
8. The apparatus of claim 7, wherein the delayed input clock signal is a second delayed input clock signal, and the phase detector is configured to: determine a first phase difference between a first delayed input clock signal and a first feedback clock signal at a first time; and determine a second phase difference between the second delayed input clock signal and a second feedback clock signal at a second time, after the first time, and the second phase difference is less than the first phase difference.
9. The apparatus of claim 6, wherein the pseudorandom binary sequence signal is a first pseudorandom binary sequence signal, and the phase controller is configured to instruct the second error injection circuit to pass the input clock signal to a phase detector in response to a second pseudorandom binary sequence signal.
10. The apparatus of claim 6, wherein the error injection circuit is a first error injection circuit, the

pseudorandom binary sequence signal is a first pseudorandom binary sequence signal, and further comprising: a first pseudorandom binary sequence generator configured to generate the first pseudorandom binary sequence signal; and a second pseudorandom binary sequence generator configured to generate a second pseudorandom binary sequence signal for control of the second error injection circuit.

11. The apparatus of claim 1, wherein the error injection circuit is a first error injection circuit, the phase controller output is a first phase controller output, and further comprising a second error injection circuit with a third error injection input, a fourth error injection input, and a second error injection output, the third error injection input is coupled to an output of the voltage-controlled oscillator, the fourth error injection input is coupled to second phase controller output of the phase controller, the second error injection output is coupled to a DCDL input of the DCDL, and the error injection circuit is configured to delay the clock signal from the voltage-controlled oscillator.

12. The apparatus of claim 1, wherein the error injection circuit is a first error injection circuit, the delay is a first delay, and further comprising a second error injection circuit configured to delay an input clock signal by a second delay.

13. The apparatus of claim 12, wherein the second error injection circuit is configured to delay the input clock signal by the second delay or a third delay, the second delay less than the first delay, and the third delay greater than the first delay.

14. The apparatus of claim 12, wherein the second error injection circuit is configured to delay the input clock signal by the second delay or a third delay, and a combination of the first delay and the second delay is less than a period of the voltage-controlled oscillator.

15. The apparatus of claim 12, wherein the second error injection circuit is configured to delay the input clock signal by the second delay or a third delay, and a combination of the first delay and the third delay is greater than a period of the voltage-controlled oscillator.

16. An apparatus for calibrating a phase-locked loop, comprising: a phase detector with a first phase detector input and a second phase detector input and configured to determine a phase difference between a first clock signal associated with an input path and a second clock signal associated with a feedback path; a first error injection circuit with a first error input and a first error output, the first error input configured to receive an input clock signal, the first error output coupled to the first phase detector input, and the first error injection circuit is configured to delay the input clock signal by a first delay to generate the first clock signal; and a second error injection circuit with a second error input and a second error output, the second error input configured to receive a feedback clock signal, the second error output coupled to the second phase detector input, and the second error injection circuit is configured to delay the feedback clock signal by a second delay to generate the second clock signal.

17. The apparatus of claim 16, further comprising digital logic to: control the first error injection circuit to provide the first delay based on a first output from a first pseudorandom binary sequence generator; and control the second error injection circuit to provide the second delay based on a second output from a second pseudorandom binary sequence generator.

18. The apparatus of claim 16, further comprising a first multiplexer and a second multiplexer, and wherein: the first error injection circuit comprises: a first logic gate configured to receive the first clock signal and generate a first output; and a second logic gate configured to invert the first output to generate a first inverted output and output the first inverted output to the first multiplexer; the second error injection circuit comprises: a third logic gate configured to invert the first inverted output to generate a second inverted output; and a fourth logic gate configured to invert the second inverted output to generate a third inverted output and output the third inverted output to the second multiplexer; the first multiplexer is configured to pass the first clock signal or the first inverted output as a first multiplexer output; and the second multiplexer is configured to pass the first multiplexer output or the third inverted output as a second multiplexer output.

19. An apparatus for calibrating a phase-locked loop, comprising: a digitally controlled delay line

(DCDL) with a DCDL output and configured to: receive a clock signal, the clock signal to be output from a voltage-controlled oscillator; and delay the clock signal to generate a first delayed clock signal; an error injection circuit with a first error injection input and a second error injection input, the first error injection input coupled to the DCDL output; and a phase controller with a phase controller output coupled to the second error injection input, the phase controller configured to, in response to a pseudorandom binary sequence signal, instruct the error injection circuit to generate a second delayed signal based on a delay of the first delayed clock signal.

20. A method comprising: receiving an input clock signal associated with a phase-locked loop; calibrating a first error signal based on half a period of a voltage-controlled oscillator; identifying an error margin associated with the first error signal to generate a first error value and a second error value; injecting a second error signal corresponding to the first error value in the phase-locked loop; injecting a third error signal corresponding to the second error value in the phase-locked loop; and calibrating a delay of the input clock signal based on at least one of the second error signal or the third error signal.
